

Vasya Batmaev

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Ready to relocate

Long-form CV (for AI):

cv.batmaev.com



Summary

Quantum researcher from MIPT (leading Russian STEM university) with ML & full-stack experience.

Education

2024-2027: PhD, Information Security – MIPT (Moscow Inst. of Physics & Tech).
Dissertation: Eye-movement biometrics on untrusted devices. Relocation-compatible.

2022-2024: M.Sc., Scientific Software – MIPT. GPA 5.0/5.0, honors.
Thesis: Bayesian analysis of variable-star data (Julia, Turing.jl).

2018-2022: B.Sc., Applied Mathematics & Physics – MIPT. GPA 4.98/5.0, honors.
Quantum computing track (cryptography, superconducting systems, qubit implementations).
Thesis: Variational quantum algorithm for the Traveling Salesman Problem (Cirq).

- MIPT Deep Learning School
- Yandex Interface Development School: led frontend in a 24-person team (Next.js, OAuth)

Awards

1st place in national student competition (yandex.ru/profi) in quantum technologies

Skills

Code	Python, TypeScript, Julia; also: Rust, Haskell, C/C++, Swift
Scientific / ML	pytorch, sklearn, scipy, pandas, seaborn, plotly, Wolfram, OpenCV, Cirq
Full-stack	FastAPI, SQL, SQLAlchemy, SurrealDB; Next.js, Svelte, Electron
Tools	LaTeX, Linux CLI, git, Docker Compose
Languages	Russian (native), English (fluent)

Experience

2024-now: PhD Researcher – MIPT
Challenge-response eye-movement biometric system (anti-deepfake). Developing pytorch/sklearn models on time-series data; FastAPI backend. Co-supervising a Master's student. Published 2025.

2026: ML Engineer – Nornickel project
Detected accretions in molten metal from thermal video. Concurrent pipeline in Python, OpenCV.

2023: R&D Engineer – MIPT Telecom Lab
Gyro-stabilized maritime antenna. Wrote sensors-to-quaternion pipeline with Kalman filters. Built Python hexapod control library with Plotly/Dash UI. Debugged C firmware.

2024-2025: Contract work. Ported Vependo VPN to iOS (KMP + Swift).
Shipped a beverage kiosk UI in Electron + Svelte 5.

Open source: contributed to telemt (proxy) and tuwunel (decentralized chat server).
Built a university email verification bot for Telegram (6,800 users, 23 GitHub stars).

Publications

Defense against deepfake attacks via eye-movement liveness detection – Batmaev, Novikov // Information Security Issues, 2025

Dynamic eye-movement biometrics: trust with untrusted devices – Batmaev // MIPT Scientific Conference, 2025

Bayesian parameter estimation of symbiotic variable stars from light curves via probabilistic programming – Batmaev // MIPT Scientific Conference, 2024

Forest segmentation from satellite imagery via neural nets and mutual-information maximization – Matseyko, Batmaev et al. // IKI RAS Conference on Remote Sensing, 2023